

2025

Booklet Number: **00227**

TEST CODE: **MTA**

Forenoon

Time: 2 hours

Read the following carefully before answering the test.

- There are 8 questions in all.
 - Questions carry equal weight.
 - Answer as many questions as you can.
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- Please write your *Registration Number, Test Centre, Test Code* and the *Number of this Question Booklet* in the designated places on the cover page of the Answer Booklet.
- Please use pens with black/blue ink to answer the questions.
- ALL ROUGH WORK MUST BE DONE ONLY ON THE ANSWER BOOKLET.
- USE OF CALCULATORS, MOBILE PHONES AND ALL TYPES OF ELECTRONIC DEVICES IS STRICTLY PROHIBITED.

STOP! WAIT FOR THE SIGNAL TO START.



1. $\mathbb{N}$ denotes the set of all positive integers.
 2. $\mathbb{Z}$ denotes the set of all integers.
 3. $\mathbb{Q}$ denotes the set of all rational numbers.
 4. $\mathbb{R}$ denotes the set of all real numbers.
 5. $\mathbb{C}$ denotes the set of all complex numbers.
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✓ 1. Show that $\lim_{N \rightarrow \infty} p(N) \int_N^\infty e^{-x^2} dx = 0$ for all polynomial $p(x)$ in $\mathbb{R}[x]$.

2. Let $\mathcal{H}$ and $\mathcal{K}$ be complex Hilbert spaces. Suppose $\{T_n\}_{n \in \mathbb{N}}$ is a sequence of bounded linear operators from $\mathcal{H}$ to $\mathcal{K}$ satisfying the following condition:

For every $\xi \in \mathcal{H}$ and for every $\eta \in \mathcal{K}$, the sequence $\{\langle T_n \xi, \eta \rangle_{\mathcal{K}}\}_{n \in \mathbb{N}}$ of complex numbers is Cauchy.

(a) Show that $\sup_{n \in \mathbb{N}} \|T_n\| < \infty$.

(b) Show that there exists a bounded linear operator $T : \mathcal{H} \rightarrow \mathcal{K}$ such that $\langle T \xi, \eta \rangle_{\mathcal{K}} = \lim_{n \rightarrow \infty} \langle T_n \xi, \eta \rangle_{\mathcal{K}}$ for every $\xi \in \mathcal{H}$, $\eta \in \mathcal{K}$.

3. Let Ω be an open subset of $\mathbb{C}$. Suppose f is a meromorphic function on Ω having **exactly one** zero at z_0 and **exactly one** pole at z_1 . Further, m is the order of zero at z_0 and n is the order of the pole at z_1 . If $g : \Omega \rightarrow \mathbb{C}$ is analytic and D is an open disc with center a and radius r such that $z_0, z_1 \in D$ and $\overline{D} \subset \Omega$, then show that

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} g(z) dz = mg(z_0) - ng(z_1)$$

where $\gamma(t) = a + re^{2\pi i t}$ for $0 \leq t \leq 1$.

4. Let $z_1, \dots, z_n \in \mathbb{C}$ such that $|z_j| = 1$ for all $j = 1, \dots, n$.

Consider the function $f : \mathbb{C} \rightarrow \mathbb{C}$ defined by

$$f(w) = (w - z_1) \cdots (w - z_n) \text{ for all } w \in \mathbb{C}.$$

Prove that the set $\{|f(w)| : |w| = 1\}$ contains the interval $[0, 1]$.

5. Let $\{A_n\}_{n \in \mathbb{N}}$ be a sequence of Borel measurable subsets of $[0, 1]$ satisfying $\lambda(A_n) = 1$ for all n (where λ denotes the Lebesgue measure on $[0, 1]$). Prove that $\bigcap_{n \in \mathbb{N}} A_n$ is dense in $[0, 1]$.

6. Let $p \in [1, \infty)$. Consider the vector space

$$\ell^p = \left\{ f : \mathbb{N} \rightarrow \mathbb{C} : \sum_{k \in \mathbb{N}} |f(k)|^p < \infty \right\}$$

with the usual norm $\|f\|_p = \left[\sum_{k \in \mathbb{N}} |f(k)|^p \right]^{\frac{1}{p}}$ for $f \in \ell^p$. Suppose $\|\cdot\|$ is another norm on ℓ^p with the following properties:

(i) $(\ell^p, \|\cdot\|)$ is complete;

(ii) If a sequence $\{f_n\}_n$ converges to f in $(\ell^p, \|\cdot\|)$, then $\lim_{n \rightarrow \infty} f_n(k) = f(k)$ for all $k \in \mathbb{N}$.

Prove that $\|\cdot\|$ is equivalent to $\|\cdot\|_p$.

7. Let $\{f_n\}_{n \in \mathbb{N}}$ be a sequence of continuous functions from $\mathbb{R}^d$ to $\mathbb{R}$, $d \geq 1$. Suppose there exist $M_1, M_\infty > 0$ and $c \in \mathbb{R}$ satisfying the following conditions:

(i) $\|f_n\|_1 \leq M_1$ and $\|f_n\|_\infty \leq M_\infty$ for all $n \in \mathbb{N}$, and

(ii) $\lim_{n \rightarrow \infty} f_n(x) = c$ for all $x \in \mathbb{R}^d$.

Prove that $c = 0$.

(Here $\|g\|_1 = \int_{\mathbb{R}^d} |g(x)| \, dx$ and $\|g\|_\infty = \sup_{x \in \mathbb{R}^d} |g(x)|$.)

8. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function which is not a polynomial. Suppose $\{P_n\}_{n \geq 1}$ is a sequence of polynomials (with real coefficients) that converges uniformly to the function f on $[0, 1]$. Prove that $\{\text{degree}(P_n) : n = 1, 2, \dots\}$ is unbounded.